Solid State Ionics 263 (2014) 146–151

Contents lists available at ScienceDirect

Solid State Ionics

journal homepage: www.elsevier.com/locate/ssi

Enhancement of high voltage cycling performance and thermal stability of $LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$ cathode by use of boron-based additives

Lingling Zhang, Yulin Ma, Xinqun Cheng, Pengjian Zuo, Yingzhi Cui, Ting Guan, Chunyu Du, Yunzhi Gao, Geping Yin *

Institute of Advanced Chemical Power Sources, School of Chemical Engineering and Technology, Harbin Institute of Technology, Harbin 150001, China

ARTICLE INFO

Article history:
Received 22 January 2014
Received in revised form 15 May 2014
Accepted 2 June 2014
Available online xxxx

Keywords:
Lithium-ion batteries
High voltage
Electrolyte additive
LiDFOB, LiBOB

ABSTRACT

The effects of lithium bisoxalatoborate (LiBOB) and lithium difluorooxalatoborate (LiDFOB) as electrolyte additives on the high voltage cycling performance of lithium-ion cells are investigated. With traces of 0.2 wt. % LiBOB or 0.5 wt. % LiDFOB added, the cycling performance of $Li/LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$ cells charged to 4.6 V is improved. The electrochemical impedance spectroscopy (EIS) suggests that the additives can decrease impedance. The results of scanning electron microscopy (SEM) of the $LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$ cathode indicated the existence of this film. X-ray photoelectron spectroscopy (XPS) indicates the additives are justified to modify cathode surface films and suppress electrolyte oxidation reactions on the cathode surface. In addition, differential scanning calorimetric (DSC) studies show that thermal stability of fully charged $Li_{1-x}Ni_{1/3}Co_{1/3}Mn_{1/3}O_2$ cathode is improved in the presence of additives.

1. Introduction

Lithium ion batteries have been widely used in electronic devices, and the demand for higher energy densities is continuously increasing [1–3]. In order to further increase the energy density of lithium ion batteries, many researchers have attempted to develop high capacity cathode materials [4–6]. $LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$ is one of the most promising alternative cathode materials for lithium-ion batteries owing to its high capacity, good structural stability and relatively low cost [7,8]. The capacity of $LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$ could be improved while the charge cut-off voltage was improved to 4.6 V. However, the conventional carbonate/$LiPF_6$ electrolyte in lithium-ion batteries tends to decompose [9,10] and the surface reactivity will increase between highly delithiated cathode material and the electrolyte at high voltage. In addition, the structure of $LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$ would change at high voltage [11]. Two methods were often used to improve the performance of high voltage cathode materials. One effective way is to coat cathode particles by metal oxides [12–15], which can inhibit the reaction between the electrolyte and the cathode surface at high voltage. Another is to develop matching electrolytes for high voltage materials [16]. Adding additives to electrolytes [17–20] is considered to be simple and effective. Lee [21] reported that electrochemically polymerized additives, such as thiophene and ethylene dioxythiophene could form a thin conductive film to suppress the electrolyte decomposition on cathode active sites at high voltage. Xu [22] reported that tris (pentafluorophenyl) phosphine formed a protective film on $LiNi_{0.5}Mn_{1.5}O_4$ to prevent electrolytes and cathode materials from oxidative decomposition. Among various additives, LiBOB or LiDFOB has been recognized as effective additives that form stable SEI on graphite electrodes to suppress the degradation of the graphite electrodes, improving the capacity retention during the cycling and aging processes at 55 °C [23]. Moreover, the cycling performance of $MCMB/Li_{1.1}[Mn_{1/3}Ni_{1/3}Co_{1/3}]_{0.9}O_2$ cell was improved using LiBOB or LiDFOB as salt at 55 °C [24]. LiBOB additive stabilized the interface between a high-voltage cathode and an electrolyte and inhibited the electrolyte decomposition, thus the high-voltage cycling performance of $Li/LiNi_{0.5}Mn_{1.5}O_4$ cells at 60 °C was improved [25]. The mechanism of the bis(oxalato)borate and difluoro(oxalate)borate anions in improving the cycling performance of lithium ion batteries was investigated in theory [26]. The solubility and ion conductivity restricted the extensive use of LiBOB, and the thick SEI layer formed in LiBOB-based electrolytes dramatically increased the interfacial impedance and deteriorated the power capability of the lithium ion batteries [27]. In addition, the irreversible capacity loss due to the formation of SEI film will also increase with increasing the amount of additive [28]. A trace additive is beneficial for reducing side effects.

In this work, the effect of trace LiBOB and LiDFOB on the cycling performance and the thermal stability of $LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$ material at high voltage were investigated by electrochemical impedance spectroscopy (EIS), X-ray photoelectron spectroscopy (XPS) and differential scanning calorimetry (DSC).

* Corresponding author. Tel./fax: +86 451 86413707.
E-mail address: yingeping@hit.edu.cn (G. Yin).

http://dx.doi.org/10.1016/j.ssi.2014.06.001
0167-2738/

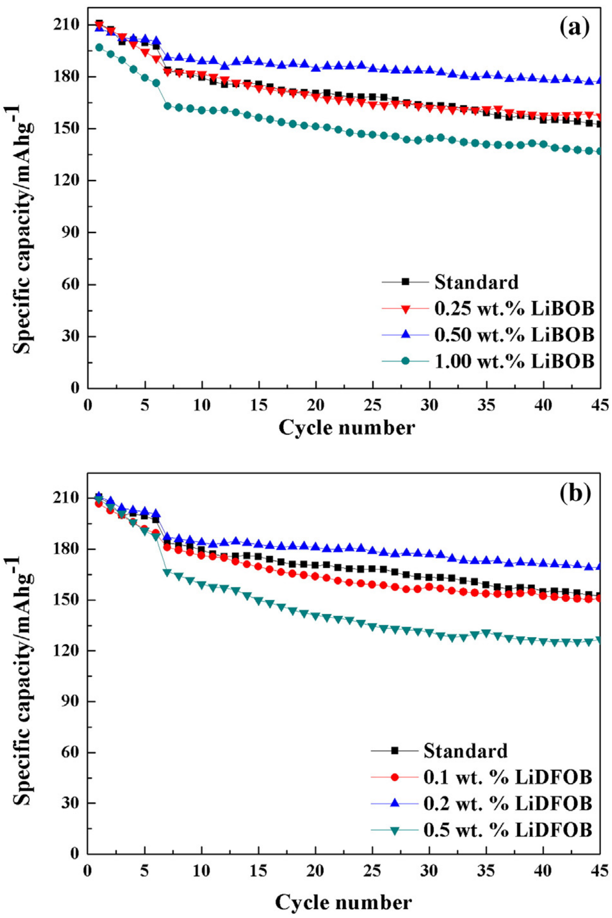

Fig. 1. Cycle performance of Li/LiNi$_{1/3}$Co$_{1/3}$Mn$_{1/3}$O$_2$ electrodes with different contents of additive in the electrolyte: (a) LiBOB (0.25 wt.%, 0.50 wt.% and 1.00 wt.%), (b) LiDFOB (0.1 wt.%, 0.2 wt.% and 0.5 wt.%).

2. Experimental

2.1. Electrode preparation and cell assembly

The standard electrolyte consisted of 1 mol L^{-1} lithium hexafluorophosphate (LiPF$_6$) salt dissolved in a mixed solvent of ethylene carbonate (EC), diethyl carbonate (DEC), and ethyl methyl carbonate (EMC) (1:1:1, by volume). The additives lithium bisoxalatoborate (LiBOB) and lithium difluorooxalatoborate (LiDFOB) (battery grade, Guangzhou Tinci Materials Technology Co., Ltd.) were used without further purification. The additives LiBOB and LiDFOB were then added into the electrolyte. The water content of electrolyte was determined by Karl Fischer titration and the value is within 15 ppm.

The electrodes were prepared by coating an N-methyl pyrrolidone (NMP)-based slurry containing LiNi$_{1/3}$Co$_{1/3}$Mn$_{1/3}$O$_2$ (3M, China), polyvinylidenefluoride (PVdF), and acetylene black at a weight ratio of 85: 5: 10 onto aluminum foil. The coin cell, which is composed of a cathode electrode LiNi$_{1/3}$Co$_{1/3}$Mn$_{1/3}$O$_2$, a metallic lithium anode and polyethylene separator (Celgard 2400), was assembled with different electrolytes in an argon-filled glove box. The reproducibility level of the capacity response for the Li/LiNi$_{1/3}$Co$_{1/3}$Mn$_{1/3}$O$_2$ cell was estimated to be approximately $\pm 3\%$, which was comparable performance in most cases.

2.2. Measurements

For the Li/LiNi$_{1/3}$Co$_{1/3}$Mn$_{1/3}$O$_2$ cells, pre-cycling was initially carried out in the potential range of 3.0–4.6 V at C/10 (1 °C = 160 mAh g^{-1}) for 6 times on a battery test instrument. After pre-cyling, charge–discharge tests were repeated for 60 cycles at room temperature (25 °C $\pm$ 2 °C) in the same potential range at 0.6 °C. The cells were allowed to rest for 2 min between charge and discharge steps.

Electrochemical impedance spectroscopy (EIS) measurements were performed using an electrochemical workstation impedance analyzer (EIS, CHI604d) over the frequency range from 1 MHz to 10 MHz with an amplitude of 10 mV.

Differential scanning calorimetry (DSC) measurements were carried out over the temperature range from 30 to 400 °C at heating rate of 10 °C min^{-1} using DSC (TG-DSC, NETZSCH, STA 449F3). The cells were fully recharged to 4.6 V after 66 cycles, and then disassembled. The LiNi$_{1/3}$Co$_{1/3}$Mn$_{1/3}$O$_2$ material was gently scraped from the current collector and sealed in closed reusable high pressure gold-plated stainless steel crucibles with the electrolyte in an argon-filled glove box. The weight ratio of electrolyte to LiNi$_{1/3}$Co$_{1/3}$Mn$_{1/3}$O$_2$ material is 1: 1.

Before surface analysis of the Li$_{1-x}$Ni$_{1/3}$Co$_{1/3}$Mn$_{1/3}$O$_2$ cathodes, the electrode was rinsed with an anhydrous DMC solvent three times to remove residual electrolyte and then vacuum-dried for 12 h at room temperature. After that, the LiNi$_{1/3}$Co$_{1/3}$Mn$_{1/3}$O$_2$ electrodes were transferred into a sample holder inside the glove box to prevent exposure to air.

Scanning electron microscopy (SEM, FEG-FEI 200 Quanta) was used to investigate the surface morphology of the LiNi$_{1/3}$Co$_{1/3}$Mn$_{1/3}$O$_2$ cathode before and after the cycling test.

Chemical compositions of the surface films on LiNi$_{1/3}$Co$_{1/3}$Mn$_{1/3}$O$_2$ electrodes were investigated by X-ray photoelectron spectroscopy (XPS, PHI-5000C ESCA system) employing AlKα radiation ($h\nu$ = 1486.6 eV) as the incident X-ray. The C1s peak at 284.6 eV was used as a reference for the final adjustment of the energy scale in the spectra. The spectra obtained were fitted using XPS peak software.

3. Results and discussions

In order to optimize the contents of the LiDFOB and LiBOB, different amounts of additive were added to the blank electrolyte. The amount of LiBOB was 0.25 wt.%, 0.50 wt.% and 1.00 wt.%, and LiDFOB was 0.1 wt.%, 0.2 wt.% and 0.5 wt.%, respectively. As shown in Fig. 1, the initial discharge capacities of the electrodes in electrolytes containing 0 wt.%, 0.25 wt.%, 0.50 wt.% and 1.00 wt.% LiBOB are 183.7 mAh g^{-1}, 183.0 mAh g^{-1}, 191.0 mAh g^{-1} and 172.8 mAh g^{-1} at 0.6 °C, respectively. The amount of LiDFOB was 0 wt.%, 0.1 wt.%, 0.2 wt.% and 0.5 wt.%, the discharge capacity was 183.7 mAh g^{-1}, 181.0 mAh g^{-1}, 187.2 mAh g^{-1} and 166.8 mAh g^{-1}. When the amount of additive was large, the initial cell performance can deteriorate. So the optimum amount was 0.2 wt.% for LiDFOB and 0.50 wt.% for LiBOB. Thus, in the

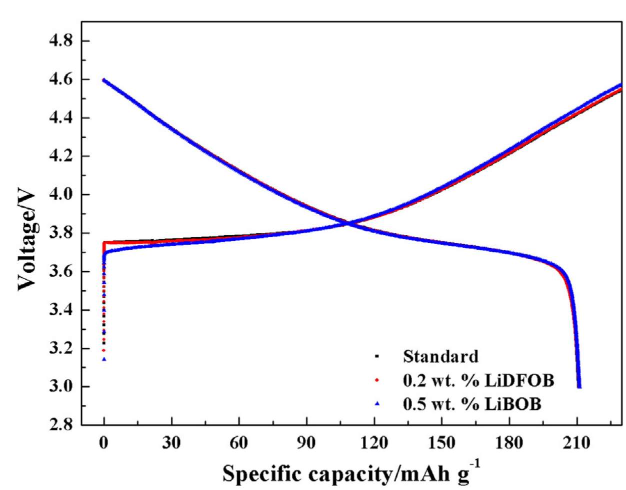

Fig. 2. First charge/discharge curves of Li/LiNi$_{1/3}$Co$_{1/3}$Mn$_{1/3}$O$_2$ cells with and without additives at 0.1 °C.

following analysis the 0.50 wt.% LiBOB or 0.2 wt.% LiDFOB is added to the electrolyte.

The reversible capacity and coulombic efficiency of the Li/$LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$ cells with/without LiBOB or LiDFOB are shown in Fig. 2. In the pre-cycling (0.1 °C), capacity and coulombic efficiency showed no significant difference between cells with standard electrolyte and optimized electrolyte (LiBOB: 208.5 mAh g^{-1}, 89.0%; LiDFOB: 211.1 mAh g^{-1}, 87.3%; without additive: 210.7 mAh g^{-1}, 86.7%), which confirmed that using LiBOB or LiDFOB did not deteriorate the initial cell performance. In order to investigate the effect of LiBOB or LiDFOB on the cycling performance of operating at high voltage, the cells with/without additives were cycled at 0.6 °C and the results are shown in Fig. 3. Using LiBOB/LiDFOB improved the cycling performance at 0.6 °C significantly. The standard cell steadily lost capacity from 183.7 mAh g^{-1} to 144.7 mAh g^{-1} with the capacity retention of 78.8% after 60 cycles, while the capacity of the cell with LiDFOB was 165.2 mAh g^{-1} after 60 cycles and retained 88.2% of its initial capacity (187.2 mAh g^{-1}). The cell with LiBOB maintained 91.8% of the initial capacity during the same cycling period. It indicated that the addition of 0.5 wt.% LiBOB or 0.2 wt.% LiDFOB could improve cycling performance at high voltage.

The electrochemical impedance spectroscopy measurements of the discharged $LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$ after 60 cycles were carried out, as shown in Fig. 4. The EIS values were fitted using the equivalent circuit shown in the inset of Fig. 4(a). The semicircle in the high frequency range was typically attributed to the resistance of a solid-state interface layer formed on the electrode surface (R_{SEI}), and the semicircle observed in the medium-to-low frequency range was ascribed to charge transfer resistance (R_{ct}) between the electrode and electrolyte [29,30]. Adding LiBOB/LiDFOB, the value of R_{SEI} decreased from 89.45 Ω to 14.21 Ω/33.1 Ω, and R_{ct} decreased from 470.5 Ω to 317.6 Ω/441.0 Ω. Compared with the standard cell, the cells containing additives exhibited lower interfacial resistance. This implies that additives were beneficial for low resistance interface layer forming on the $LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$ during the high operating voltage. We can observe that the cell without additives had the largest charge transfer resistance. Owing to electrolyte decomposition, the passivation layer formed on the electrode surface, which may hamper charge transfer through the interface between electrode and electrolyte, resulting in increasing the charge transfer resistance. It is consistent with the improved capacity retention and efficiency at lower concentrations of LiBOB or LiDFOB.

Fig. 5 showed the SEM images of the $LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$ cathode surfaces before and after 60 cycles in different electrolytes. The surface of $LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$ cathode after 60 cycles in additive-containing electrolyte is obviously different from that in a standard electrolyte,

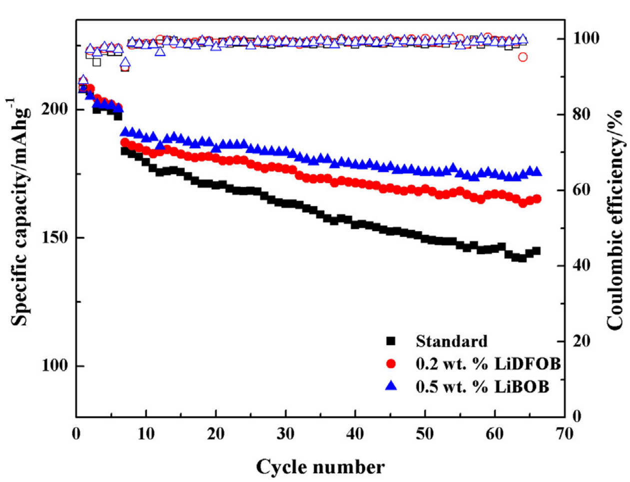

Fig. 3. Reversible capacities and efficiencies of cells with and without additives.

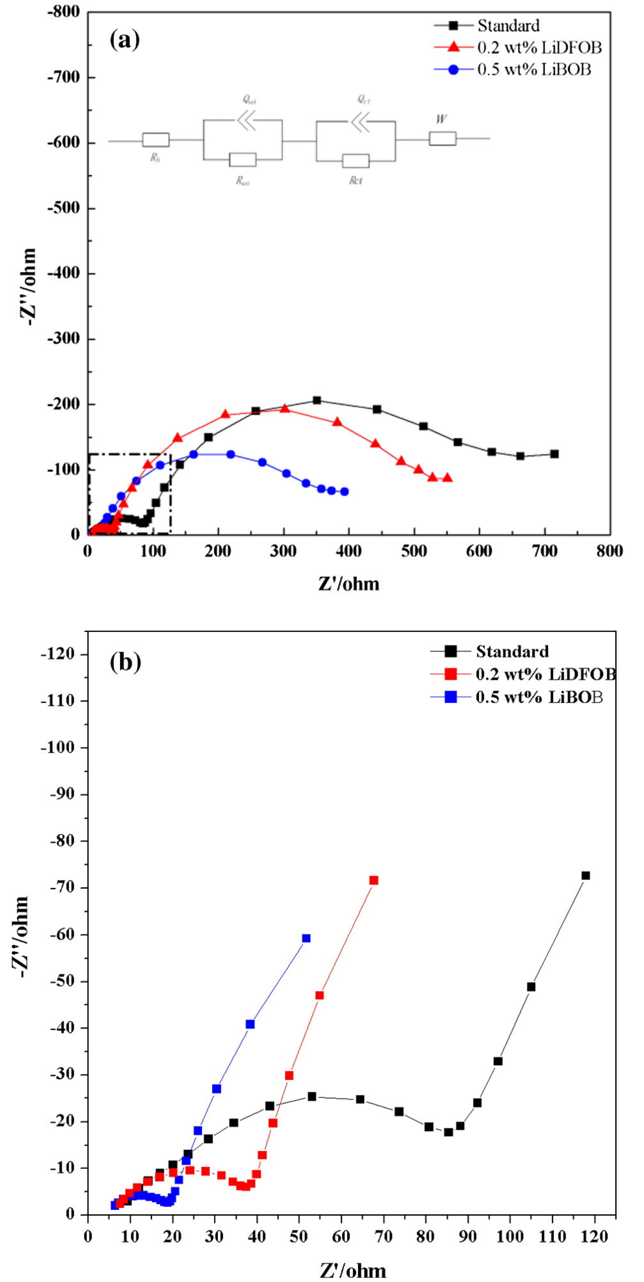

Fig. 4. (a) EIS of cells with/without LiBOB or LiDFOB (b) partial enlarged drawing of Fig. 4a.

demonstrating that additive is an effective agent for forming the surface film.

XPS has been used to study surface films, which can analyze the top ~5.0 nm of the surface. It has good sensitivity for the elements of interest (C, O, F, and P), and can determine chemical bonding states (C–O, C═O, C–C, etc.) [31].

To further study the composition of surface films on an $LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$ electrode, XPS was performed using fully charged electrodes after 60 cycles. XPS spectra of the fresh and cycled $LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$ cathodes were provided in Fig. 6. From the C1s spectra, the fresh cathode was mainly composed of peaks at C–H (285.7 eV), C–F (290.4 eV) and C–C (284.3 eV), belonging to PVDF and carbon absorbed from the atmosphere. New peak characteristics of C–O (286–287 eV) and C═O (288–289 eV) were detected on the cycled electrode surface which was due to the decomposition of electrolyte on the cathode surface. The intensity of the C–O and C═O peaks was weaker for cells cycled with additives than cells cycled with the standard electrolyte,

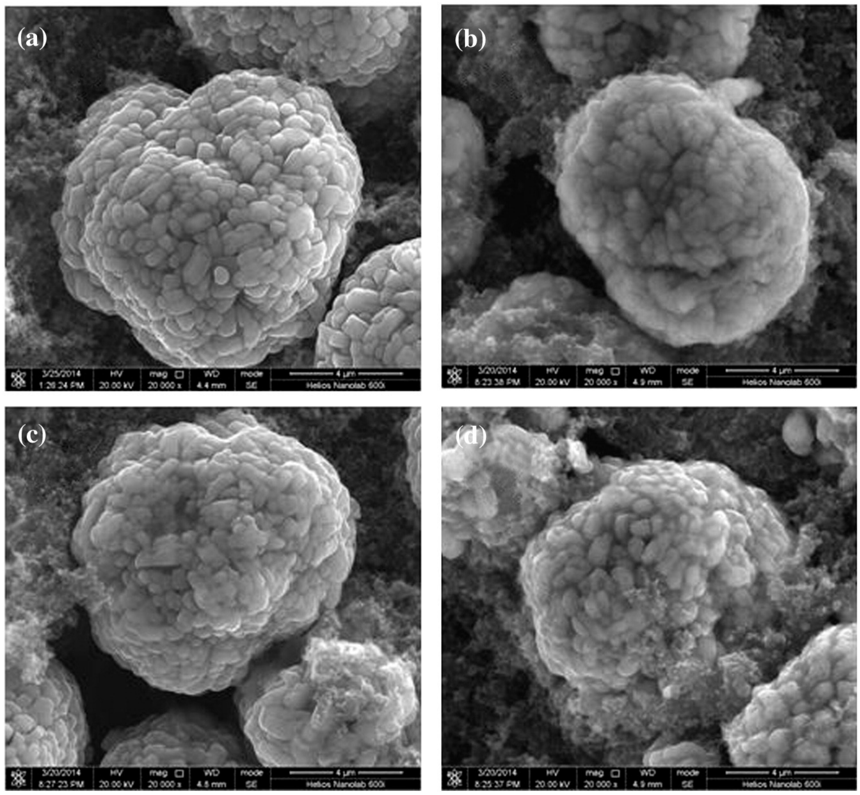

Fig. 5. SEM images of the $LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$ electrodes before and after 60 cycles in different electrolyte: (a) before cycle, (b) standard electrolyte, (c) 0.5 wt.% LiBOB, (d) 0.2 wt.%LiDFOB.

which is due to less organic degradation products, including poly(ethylene carbon-ate) (PEC), lithium alkyl carbonates, oxalates, and related electrolyte decomposition products, covering the surface of the cathode electrode [32–34].

Differences can also be observed in O1s spectra. The fresh cathode surface was mainly composed of metal oxide (529.5 eV) and Li_2CO_3 (531.5 eV) [34]. Compared with fresh cathode, the intensity of metal oxide (529.5 eV) was much lower for cycled cathode. Furthermore,

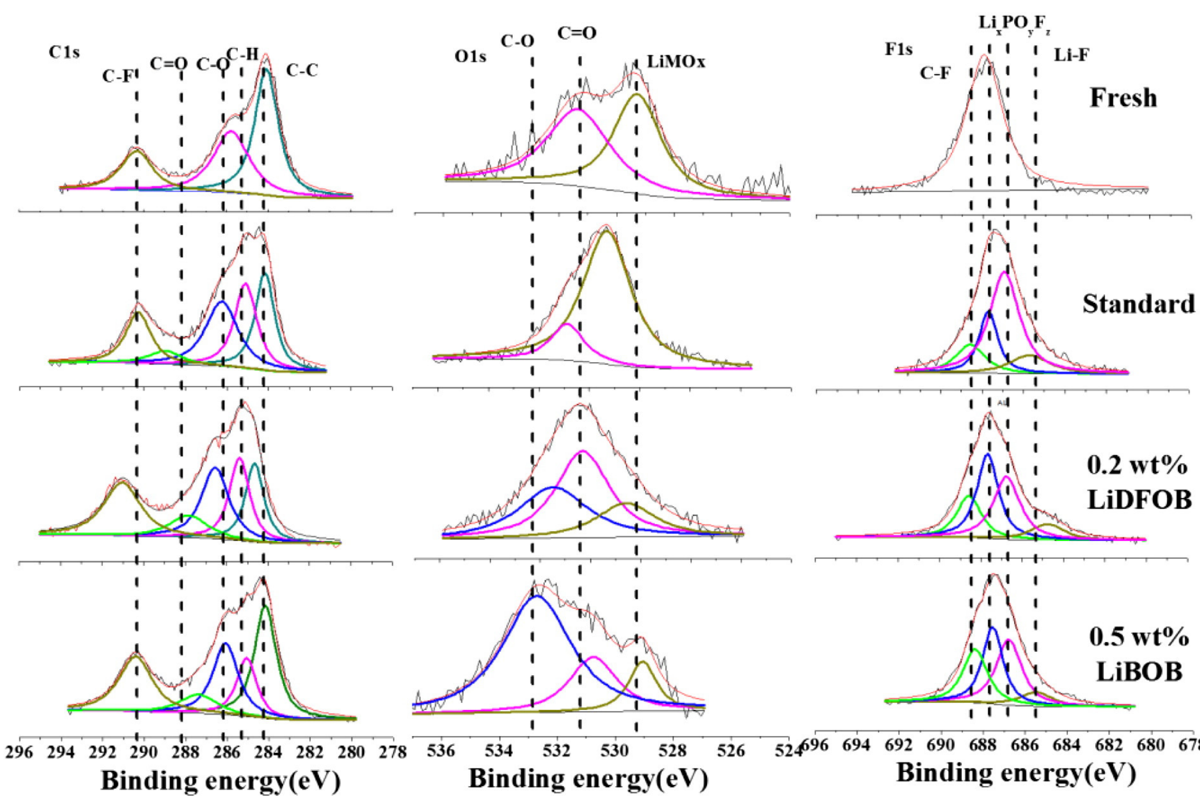

Fig. 6. XPS spectra of the fresh and cycled cathodes with/without additives.

the intensity of metal oxide (529.5 eV) was much stronger for cells cycled with additives than that of a standard electrolyte. This indicated that the cathode was covered with a thick surface film in a standard electrolyte.

The F1s spectra were fitted with four peaks centered at 689 eV (C−F), 687–688 eV (reduction products of $LiPF_6$) and 685.5 eV (Li-F) [3,35]. From the F1s spectra, the cathodes extracted from cells cycled with electrolytes containing additives had small peak characteristic of LiF at 685 eV, than that of the cell containing standard electrolyte.

From the B1s XPS of the cathodes, the cell cycled with additive contained a broad trigonal-borate signal at 192 to 193 eV (not shown here) which supported the presence of borates [36]. The presence of borates on the surface of the cathodes indicated the oxidation of the additives on the cathode surface, which might provide a protective passivation layer preventing further oxidation of the electrolyte.

In order to understand the influence of the additive on Li electrode in $Li/LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$ cells, a galvanostatic cycling test was carried out using a Li/Li symmetrical battery. The potential change for Li stripping and deposition in Li/Li symmetrical cells with and without an additive at 0.2 °C and 0.6 °C, is shown in Fig. 7. Compared with the standard electrolyte, potential drops at 0.6 °C were slightly increased in LiBOB/LiDFOB-containing electrolyte, indicating that the additive is less effective to enhance the galvanostatic cycling of Li/Li symmetrical cells [25]. Therefore, the improved high voltage performance of a $Li/LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$ battery is closely linked to a thinner protective film on the cathode.

The thermal stability of cathode materials, especially in the delithiated state, is very important for battery safety. In order to evaluate the effect of additive for the thermal stability of the delithiated $LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$ material, DSC measurements were performed. Fig. 8 shows the DSC profiles of the cathode materials charged to 4.6 V. The exothermic peak around 230 °C can be attributed to the decomposition of an electrolyte caused by an active cathode surface, and the peak around 260 °C can be attributed to the electrolyte oxidation caused by released oxygen from $Li_{1-x}Ni_{1/3}Co_{1/3}Mn_{1/3}O_2$ [37,38]. The $Li_{1-x}Ni_{1/3}Co_{1/3}Mn_{1/3}O_2$ material in the cell with LiBOB/LiDFOB additive had a smaller exothermic heat at higher temperature, which suggested that the surface reaction was inhibited owing to the thinner film. The exothermic peaks were shifted to lower temperature, and the heat generation was reduced. The total heat was reduced from 607.9 J g^{-1} to 547.4 J g^{-1}/549.9 J g^{-1} by adding LiBOB or LiDFOB, which indicated that the additive could enhance thermal stability of the delithiated cathode material to some extent. We believe that the layer coated on cathode surface may block the oxygen release and it can shift the exothermic oxidation reaction to a higher temperature as well as reduce the heat generation. These results suggest that the thermal stability of a lithium ion battery can be improved by adding a small amount of organic additive into the liquid electrolyte.

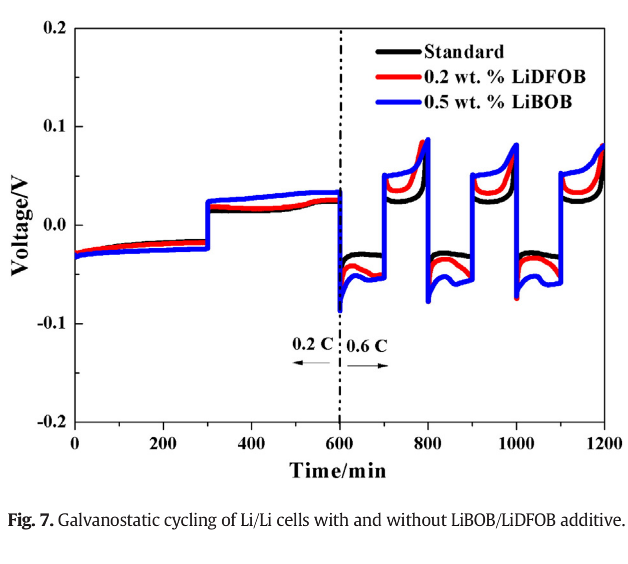

Fig. 7. Galvanostatic cycling of Li/Li cells with and without LiBOB/LiDFOB additive.

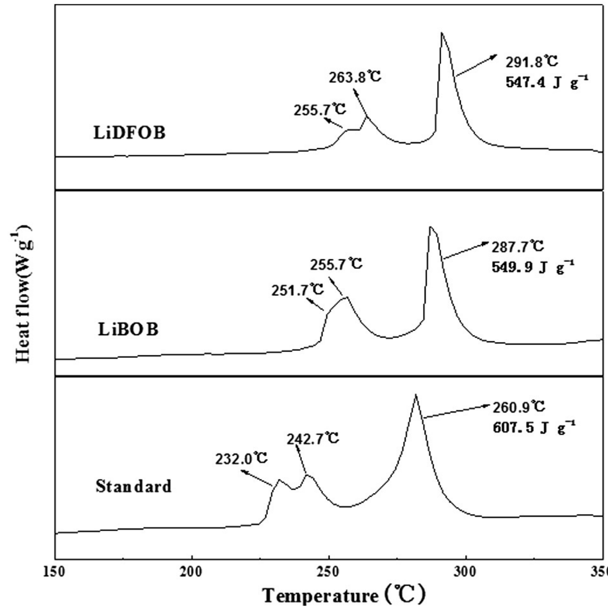

Fig. 8. DSC profiles of $Li_{1-x}Ni_{1/3}Co_{1/3}Mn_{1/3}O_2$ cathode materials charged to 4.6 V after 60 repeated cycles.

4. Conclusion

The effect of trace of LiBOB/LiDFOB on the electrochemical and thermal behavior of the $Li/LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$ cell was studied in this paper. The additive participates in the formation of film on the surface of a $LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$ cathode, suppressing the decomposition of EC, and the cycling performance and thermal stability of high-voltage lithium-ion cells were improved. The additive LiBOB/LiDFOB is promising to promote the high-voltage application in lithium ion batteries. We expect that the results of this study and the associated analysis will contribute to develop high-voltage electrolyte.

Acknowledgments

This work was funded by the National High Technology Research and Development Program (863 Program) of China (no. 2012AA110203), the National Science Foundation of China (no. 51202047), the Heilongjiang Postdoctoral Fund (no. LBH-Z11141), and the Fundamental Research Funds for the Central Universities (no. HIT. NSRIF. 2011022).

References

[1] K. Xu, Chem. Rev. 104 (2004) 4303–4417.
[2] B. Dunn, H. Kamath, J.M. Tarascon, Science 334 (2011) 928–935.
[3] L.X. Liao, T. Fang, X.G. Zhou, Y.Z. Gao, X.Q. Cheng, L.L. Zhang, G.P. Yin, Solid State Ionics 254 (2014) 27–31.
[4] X. Sun, C.A. Angell, Electrochem. Commun. 11 (2009) 1418–1421.
[5] A. Abouimrane, I. Belharouak, K. Amine, Electrochem. Commun. 11 (2009) 1073–1076.
[6] A. Manthiram, J. Phys. Chem. 2 (2011) 176–184.
[7] M.H. Lee, Y.J. Kang, S.T. Myung, Y.K. Sun, Electrochim. Acta 50 (2004) 939–948.
[8] W. Lu, Z. Chen, H. Joachin, J. Prakash, J. Liu, K. Amine, J. Power Sources 163 (2007) 1074–1079.
[9] K.M. Shaju, G.V. Subba Rao, B.V.R. Chowdari, Electrochim. Acta 48 (2002) 145–151.
[10] Y.K. Sun, S.W. Cho, S.W. Lee, C.S. Yoon, K. Amine, J. Electrochem. Soc. 154 (2007) A168–A172.
[11] H. Wang, Y.I. Jang, B. Huang, D.R. Sadoway, Y.M. Chiang, J. Electrochem. Soc. 146 (1999) 473–480.
[12] S.J. Shi, J.P. Tu, Y.Y. Tang, Y.Q. Zhang, X.Y. Liu, X.L. Wang, C.D. Gu, J. Power Sources 225 (2013) 338–346.

[13] X.Z. Liu, H.Q. Li, E. Yoo, M. Ishida, H.S. Zhou, Electrochim. Acta 83 (2012) 253–258.
[14] A.M.A. Hashem, A.E. Abdel-Ghany, A.E. Eid, J. Trottier, K. Zaghib, A. Mauger, C.M. Julien, J. Power Sources 196 (2011) 8632–8637.
[15] C.X. Ding, Y.C. Bai, X.Y. Feng, C.H. Chen, Solid State Ionics 189 (2011) 69–73.
[16] L.L. Zhang, Y.L. Ma, C.Y. Du, G.P. Yin, Process. Chem. 26 (4) (2013) 553–559.
[17] R. Mogi, M. Inba, S. Jeong, Y. Iriyama, T. Abe, Z. Ogumi, J. Electrochem. Soc. 149 (2002) A1578–A1583.
[18] H. Ota, K. Shima, M. Ue, J. Yamaki, Electrochim. Acta 49 (2004) 565–572.
[19] Y.K. Han, J. Jung, S. Yu, H. Lee, J. Power Sources 187 (2009) 581–585.
[20] K. Abe, Y. Ushigoe, H. Yoshitake, M. Yoshio, J. Power Sources 153 (2006) 328–335.
[21] Y.S. Lee, K.S. Lee, Y.K. Sun, Y.M. Lee, D.W. Kim, J. Power Sources 196 (2011) 6997–7001.
[22] M.Q. Xu, Y.L. Liu, B. Li, W.S. Li, X.P. Li, S.J. Hu, Electrochem. Commun. 18 (2012) 123–126.
[23] I. Belharouak, K. Amine, J. Power Sources 174 (2007) 852–855.
[24] Z. Chen, J. Liu, K. Amine, Electrochem. Solid-State Lett. 10 (2007) A45–A47.
[25] S.Y. Ha, J.G. Han, Y.M. Song, M.J. Chun, S.I. Han, W.C. Shin, N.S. Choi, Electrochim. Acta 104 (2013) 170–177.
[26] Ilya A. Shkrob, Ye Zhu, Timothy W. Marin, Daniel P. Abraham, J. Phys. Chem. C 117 (2013) 23750–23756.
[27] K. Xu, S.S. Zhang, R. Jow, J. Power Sources 143 (2005) 197.
[28] S. Dalavi, M.Q. Xu, B. Knight, B.L. Lucht, Electrochem. Solid-State Lett. 15 (2) (2012) A28–A31.
[29] A. Funabiki, M. Inaba, Z. Ogumi, J. Power Sources 68 (1997) 227–231.
[30] M.D. Levi, G. Salitra, B. Markovsky, H. Teller, D. Aurbach, U. Heider, L. Heider, J. Electrochem. Soc. 146 (1999) 1279–1289.
[31] M. Lu, H. Cheng, Y. Yang, Electrochim. Acta 53 (2008) 3539–3546.
[32] H. Duncan, Y. Abu-Lebdeh, I.J. Davidson, J. Electrochem. Soc. 157 (2010) A528–A535.
[33] L. Yang, B. Ravdel, B.L. Lucht, Electrochem. Solid-State Lett. 13 (2010) A95–A97.
[34] R. Dedryvere, D. Foix, S. Franger, S. Patoux, L. Daniel, D. Gonbeau, J. Phys. Chem. C 114 (2010) 10999–11008.
[35] A.M. Andersson, K. Edström, J. Electrochem. Soc. 148 (2001) A1100–A1109.
[36] K. Xu, U. Lee, S.S. Zhang, M. Wood, T.R. Jow, Electrochem. Solid-State Lett. 6 (2003) A144–A148.
[37] Y. Baba, S. Okada, J. Yamaki, Solid State Ionics 148 (2002) 311–316.
[38] J. Li, Z.R. Zhang, X.J. Guo, Y. Yang, Solid State Ionics 177 (2006) 1509–1516.